

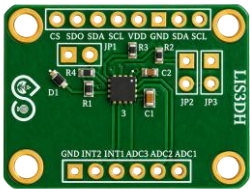
The LIS3DH is an ultra-low-power high performance three-axis linear accelerometer belonging to the “nano” family, with digital I2C/SPI serial interface standard output.

The device may be configured to generate interrupt signals using two independent inertial wake-up/free-fall events as well as by the position of the device itself. Thresholds and timing of interrupt generators are programmable by the end user on the fly.

LIS3DH has an integrated 32-level first-in, first-out (FIFO) buffer allowing the user to store data in order to limit intervention by the host processor.

Key Features

- Ultra-low power 3-axis accelerometer — ideal for battery-powered devices
- Multiple motion detection functions — free-fall, tap/click, activity, and orientation
- Flexible performance — supports both high-resolution and low-power modes
- Built-in FIFO buffer — reduces processor load and saves power
- Compact LGA package — small size (3 × 3 × 1 mm) for wearables and mobile devices



LIS3DH Sensor

Technical Specifications

Parameter	Value
Supply Voltage	1.71 V to 3.6 V (wide supply voltage range)
Operating Temperature Range	−40 °C to +85 °C
IO Supply	Independent IO supply (1.8 V) and supply voltage compatible
Ultra-Low-Power Mode Consumption	Down to 2 μA
Digital Interface	I2C/SPI digital output interface
Data Output	16-bit data output
Interrupt Generators	2 independent programmable interrupt generators for free-fall and motion detection
Free-Fall Detection	Supported
Motion Detection	Supported
FIFO Buffer	Embedded 32 levels of 16-bit data output FIFO
Shock Survivability	10000 g high shock survivability
Measurement Range (selectable)	±2 g / ±4 g / ±8 g / ±16 g

I2C Logic Pins

- The I2C Address Jumper — Bridge to use alternate address 0x18, otherwise leave open for 0x19. Leave open for SPI use.

- The I2C Pull-up Enable — Closed by default, this connects a pull-up resistor between the I2C lines and VCC. This generally doesn't interfere with SPI operation, but, if less power consumption is required, carefully cut the copper traces.

Applications

- Motion activated functions
- Free-fall detection
- Display orientation
- Gaming and virtual reality input devices
- Impact recognition and logging
- Vibration monitoring and compensation